

Machine Learning MCQs (50 Questions)

Module 1: Introduction to Machine Learning

1. Machine Learning is a subset of:

- A) Computer Networks
- B) Artificial Intelligence
- C) Database Systems
- D) Operating Systems

2. Which type of learning uses labeled data?

- A) Unsupervised Learning
- B) Reinforcement Learning
- C) Supervised Learning
- D) Semi-supervised Learning

3. Which of the following is an example of a regression problem?

- A) Spam Detection
- B) Disease Classification
- C) House Price Prediction
- D) Image Recognition

4. Which of the following is an example of a classification problem?

- A) Temperature Prediction
- B) Stock Price Forecasting
- C) Email Spam Detection
- D) Sales Forecasting

5. Overfitting occurs when a model:

- A) Performs well on training and test data
- B) Learns noise from training data
- C) Has very few features
- D) Cannot learn patterns

6. Underfitting occurs when:

- A) The model is too complex
- B) The model perfectly fits data
- C) The model fails to capture underlying patterns
- D) The dataset is very large

7. The primary purpose of a cost function is to:

- A) Increase accuracy
- B) Measure prediction error
- C) Store data
- D) Reduce features

8. Which dataset is used to train a machine learning model?

- A) Validation Set
 - B) Test Set
 - C) Training Set
 - D) Random Set
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Linear Regression

9. Linear Regression is mainly used for:

- A) Clustering
- B) Classification
- C) Regression
- D) Dimensionality Reduction

10. The equation of simple linear regression is:

- A) $y = mx + c$
- B) $y = x^2$
- C) $y = \log(x)$
- D) $y = e^x$

11. Which assumption states that the relationship between variables is linear?

- A) Homoscedasticity
- B) Independence
- C) Linearity
- D) Normality

12. The slope in linear regression represents:

- A) Error
- B) Rate of change of y with x
- C) Intercept
- D) Cost

13. Which method is commonly used to find the best-fit line?

- A) K-Means
 - B) Least Squares Method
 - C) BFS
 - D) DFS
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Logistic Regression

14. Logistic Regression is mainly used for:

- A) Regression
- B) Clustering
- C) Classification
- D) Forecasting

15. Which function is used in Logistic Regression?

- A) ReLU
- B) Linear Function
- C) Sigmoid Function
- D) Tanh Function

16. The output range of the sigmoid function is:

- A) $(-\infty, \infty)$
- B) $[0, 1]$
- C) $[-1, 1]$
- D) $[1, 10]$

17. Logistic Regression predicts:

- A) Distances
- B) Probabilities
- C) Clusters
- D) Features

18. Odds ratio in Logistic Regression is:

- A) $p/(1-p)$
 - B) $p \times (1-p)$
 - C) $1/p$
 - D) p^2
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K-Nearest Neighbors (KNN)

19. KNN is known as a:

- A) Eager Learner
- B) Lazy Learner
- C) Deep Learner
- D) Online Learner

20. KNN is a:

- A) Parametric Algorithm
- B) Non-Parametric Algorithm
- C) Probabilistic Algorithm
- D) Optimization Algorithm

21. Which distance measure is most commonly used in KNN?

- A) Manhattan Distance
- B) Euclidean Distance
- C) Hamming Distance
- D) Cosine Distance

22. Why is an odd value of K often preferred?

- A) Faster execution
- B) Reduces memory usage
- C) Avoids ties in voting
- D) Improves storage

23. During prediction, KNN:

- A) Builds a decision tree
- B) Finds nearest neighbors
- C) Creates clusters
- D) Uses gradient descent

24. KNN can be used for:

- A) Classification only
- B) Regression only
- C) Both Classification and Regression
- D) Clustering only

Naïve Bayes Classifier

25. Naïve Bayes is based on:

- A) Bayes Theorem
- B) Pythagoras Theorem
- C) Newton's Law
- D) Markov Property

26. The key assumption of Naïve Bayes is:

- A) Feature Dependence
- B) Feature Independence
- C) Equal Variance
- D) Equal Means

27. Conditional probability is represented as:

- A) $P(A)$
- B) $P(B)$
- C) $P(A|B)$
- D) $P(A+B)$

28. Which Naïve Bayes variant is commonly used for continuous features?

- A) Bernoulli Naïve Bayes
- B) Multinomial Naïve Bayes
- C) Gaussian Naïve Bayes
- D) Categorical Naïve Bayes

29. Naïve Bayes is generally:

- A) Slow
 - B) Computationally Efficient
 - C) Unstable
 - D) Non-probabilistic
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Decision Tree

30. A Decision Tree is primarily used for:

- A) Classification and Regression
- B) Clustering
- C) Association Rules
- D) Compression

31. Information Gain is based on:

- A) Entropy
- B) Variance
- C) Mean
- D) Distance

32. The attribute with the highest Information Gain is selected as:

- A) Leaf Node
- B) Root Node
- C) Child Node
- D) Decision Boundary

33. Gini Index measures:

- A) Purity of a node
- B) Distance
- C) Accuracy
- D) Variance

34. Lower Gini Index indicates:

- A) Higher Impurity
 - B) Better Purity
 - C) More Features
 - D) Larger Dataset
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Support Vector Machine (SVM)

35. The main objective of SVM is to:

- A) Create clusters
- B) Maximize margin between classes
- C) Reduce dimensions
- D) Increase features

36. The separating boundary in SVM is called:

- A) Centroid
- B) Hyperplane
- C) Cluster
- D) Node

37. Support vectors are:

- A) Random points
- B) Closest points to the hyperplane

- C) Outliers
- D) Cluster centers

38. Linear SVM is suitable when:

- A) Data is linearly separable
- B) Data has missing values
- C) Data is unlabeled
- D) Data is clustered

39. Which technique helps SVM solve non-linear problems?

- A) Pruning
- B) Bootstrapping
- C) Kernel Trick
- D) Sampling

40. Which kernel is commonly used in non-linear SVM?

- A) Linear Kernel
 - B) RBF Kernel
 - C) Mean Kernel
 - D) Average Kernel
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Confusion Matrix

41. A Confusion Matrix is used to evaluate:

- A) Regression Models
- B) Classification Models
- C) Clustering Models
- D) Optimization Models

42. True Positive (TP) means:

- A) Predicted Positive and Actually Positive
- B) Predicted Positive and Actually Negative
- C) Predicted Negative and Actually Positive
- D) Predicted Negative and Actually Negative

43. False Positive (FP) means:

- A) Correct Positive Prediction
- B) Incorrect Positive Prediction
- C) Correct Negative Prediction
- D) Incorrect Negative Prediction

44. Sensitivity is also known as:

- A) Precision
- B) Recall
- C) Accuracy
- D) Specificity

45. Precision measures:

- A) Correct Positive Predictions among Predicted Positives
- B) Correct Negative Predictions
- C) Overall Accuracy
- D) Error Rate

46. Misclassification Rate is:

- A) Correct Predictions / Total Predictions
 - B) Incorrect Predictions / Total Predictions
 - C) $TP / (TP + FN)$
 - D) $TN / (TN + FP)$
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Module 4: Advanced Topics & ANN

47. Semi-supervised learning uses:

- A) Only labeled data
- B) Only unlabeled data
- C) Both labeled and unlabeled data
- D) Reward signals only

48. Reinforcement Learning learns through:

- A) Labels
- B) Clustering
- C) Rewards and Penalties
- D) Sampling

49. In an Artificial Neural Network (ANN), the layer that receives input data is called:

- A) Hidden Layer
- B) Output Layer
- C) Input Layer
- D) Activation Layer

50. The primary role of an activation function in ANN is to:

- A) Store data
 - B) Introduce non-linearity
 - C) Remove neurons
 - D) Reduce dataset size
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Time Series (Bonus Coverage)

51. Seasonal variations in a time series occur:

- A) Randomly
- B) At regular intervals
- C) Once in a lifetime
- D) Due to errors only

52. Cyclic variations generally occur over:

- A) Short fixed periods
- B) Long-term business cycles
- C) Daily intervals
- D) Weekly intervals

53. Seasonal Index is used to measure:

- A) Trend
- B) Seasonal Effect
- C) Random Error
- D) Correlation

54. Time series data consists of observations:

- A) Collected randomly
- B) Collected over time
- C) Collected only once
- D) Collected from clusters

55. Which of the following is a component of a time series?

- A) Trend
- B) Seasonal Variation
- C) Cyclic Variation
- D) All of the Above